American Computer Science League

2024-2025 ● Contest 4: Short Problems Solutions ● Intermediate Division

1. Graph Theory

The adjacency matrix raised to the 2nd power for paths of length 2 is as follows:

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

There are 8 entries of 1.

C. 8

2. Graph Theory

There are 11 cycles in the graph. They are: ABA, ADA, AEBA, AEDA, ABEDA, ADCEBA, ABCEDA, BEB, BCEB, CEC, CEDC.

D. 11

3. Digital Electronics

This circuit translates to the Boolean expression: $\overline{\overline{A(AB)} + \overline{B + C}}$

$$\begin{aligned} \overline{\overline{A(AB)} + \overline{B + C}} &= \overline{\overline{AB} + \overline{B + C}} \\ &= \overline{\overline{AB}}\,\overline{\overline{B + C}} \\ &= AB(B + C) \\ &= AB + ABC \\ &= AB(1 + C) \\ &= AB \end{aligned}$$

This is TRUE for (1,1,*). Therefore 2 triples make it TRUE

B. 2

4. Digital Electronics

This translates to : $\overline{(A + B) \oplus BC + \overline{C}}$ since XOR is a higher priority than OR.

E. $\overline{(A + B) \oplus BC + \overline{C}}$

5. Assembly Language

```
NUM    DC       2025
       LOAD     NUM
       DIV      =100
       STORE    A
       MULT     =100
       SUB      NUM
       MULT     =-1
       STORE    B
       DIV      =5
       MULT     =10
       STORE    ANS
       LOAD     A
       DIV      =5
       ADD      ANS
       STORE    ANS
       PRINT    ANS
       END
```

This program contains only sequential statements that store the first 2 digits of 2025 in the variable A (20) and the last 2 digits of the 2025 in the variable B (25). ANS becomes 25/5 and multiplies it by 10 to make it the tens digit before adding 20/5 so the answer is 5*10+4=54, thus reversing the values of 25/5 and 20/5.

B. 54

6. Assembly Language

```
T      DC       0
N      DC       19
NUM    DC       135
TOP    LOAD     NUM
       SUB      T
       BE       STOP
       LOAD     N
       ADD      =1
       STORE    N
       LOAD     T
       ADD      N
       STORE    T
       BU       TOP
STOP   PRINT    N
       END
```

This program finds the sum of sequential numbers adding up to 135 and prints the last value that is needed to equal 135. The sum 20 + 21 + 22 + 23 + 24 + 25 = 135 so the answer is 25.

This problem was inspired by the fact that 179 + 180 + … + 189 = 2024.

E. 25